

CS7DS2 Final Assignment: Method Performance Summary

Q1: Adaptive Step Size (120 iters)		Best Bench C:
Method	Final f (A / B / C)	Q1: Polyak (0.024) Q2: Nesterov (0.479) Q3: Newton (0.686)
GD baseline	0.4826 / 0.7244 / 3.514	Key insight: kappa(A)=1.12 -> trivial kappa(C)=2508 -> hard! All methods converge A,B
Polyak (f*=0)	4.767 / 0.982 / 0.024	
Adagrad	0.4826 / 0.7244 / 2.203	
RMSprop	0.506 / 0.728 / 3.120	
Heavy Ball	0.4826 / 0.7244 / 1.400	
Q2: Momentum & SGD (150 iters)		Huge spread on C
Method	Final f (A / B / C)	
Nesterov	0.4826 / 0.7244 / 0.479	
Adam	0.4826 / 0.7244 / 1.579	
SGD (b=5)	~0.483 / N/A / N/A	
SGD (b=40)	~0.483 / N/A / N/A	
Q3: 2nd Order (Newton 20 / GD 80)		
Method	Final f (A / B / C)	
Newton	0.4826 / 0.7244 / 0.686	
GD baseline	0.4827 / 0.7244 / 3.732	
Q4: Derivative-Free (various)		
Method	Final f (A / B / C)	
FD delta=0.05	similar to GD	
FD delta=0.8	suboptimal	
Zeroth-order	slow convergence	
Nelder-Mead	B/C only	